IronCircle

Certification Report

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Final Assessment Report Submission

Case: Mail n' Trail

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1. Executive Summary

This investigation was initiated after suspicious download activity was detected by a Cowrie honeypot deployed within the organization's monitoring environment. The purpose of the investigation was to identify the nature of the activity, determine whether attacker behavior indicated payload staging or command-and-control preparation, and reconstruct the full sequence of events using SIEM telemetry.

Due to limited visibility into the monitoring platform, the analyst first restored investigative access by pivoting through an internal mail server to recover Splunk credentials. Following SIEM access, a hypothesis-driven threat hunting approach was used to identify command execution patterns, correlate attacker infrastructure usage, retrieve staged payload artifacts, and perform decoding analysis.

The investigation confirmed repeated attacker interaction with the honeypot environment involving external payload retrieval using command-line tooling. This behavior represents realistic adversary techniques including ingress tool transfer, infrastructure staging, and payload obfuscation.

2. Investigation Context

Cowrie is an SSH/Telnet honeypot designed to simulate vulnerable systems in order to capture attacker behavior. The monitoring system recorded abnormal download activity suggesting that an adversary may have gained access to the honeypot environment and attempted to retrieve external content. In real-world scenarios, such activity may indicate malware staging, persistence preparation, or reconnaissance.

3. Investigation Methodology and Analytical Reasoning

3.1 Restoring Visibility Through Internal Pivoting

The SIEM platform containing honeypot telemetry was initially inaccessible due to missing credentials. To restore investigative capability, the analyst connected to an internal POP3 mail server using provided credentials.

```
Telnet 10.0.69.93
+OK Dovecot (Ubuntu) ready.
User
-ERR Unknown command.

-ERR Unknown command.
USER johnd
+OK
PASS toor
+OK Logged in.
LIST
+OK 4 messages:
1 1117
2 1361
3 1210
4 869
.
```

Figure 1 — Successful authentication and mailbox enumeration on internal POP3 server.

This step represents a common real-world DFIR workflow where investigators must first recover monitoring access before analyzing threat data. Mailbox enumeration revealed operational communications including administrative credentials for the Splunk platform.

```
Telnet 10.0.69.93
Received: by splunk-server.ec2.internal (Postfix, from userid 0)
        id 914CC1B9364; Mon, 19 Aug 2024 10:30:50 +0000 (UTC)
Subject: We launched Splunk!
Message-Id: <20240819103050.914CC1B9364@splunk-server.ec2.internal>
Date: Mon, 19 Aug 2024 10:30:50 +0000 (UTC)
From: root <root@corporate.com>

Hey there,

We wanted to call your attention to our new SIEM product we released overnight.
Using this new feature, you'll be able to investigate for suspicious events.
If you have any questions about the best ways to use Splunk, please feel free to give us a call at +1-202-
555-0128.

To use Splunk, navigate to the following URI:
http://[SERVER-IP]:8000/

The credentials are as follows:
username: admin
password: CTF_Final!

Thank you,
Admin

.
```

Figure 2- Administrative email containing Splunk access credentials.

3.2 Dataset Familiarization and Log Normalization Challenge

After gaining access to Splunk, broad exploratory searches were executed to understand log volume and data structure. It was discovered that Cowrie logs were ingested as CSV-formatted syslog entries rather than normalized honeypot sourcetypes. This required manual behavioral hunting rather than reliance on structured detection fields, significantly increasing analytical complexity.

3.3 Hypothesis-Driven Threat Hunting

The analyst formed a hypothesis that suspicious download activity would likely involve command-line tools commonly used by attackers for payload retrieval. Keyword searches targeting terms such as 'curl' were performed.

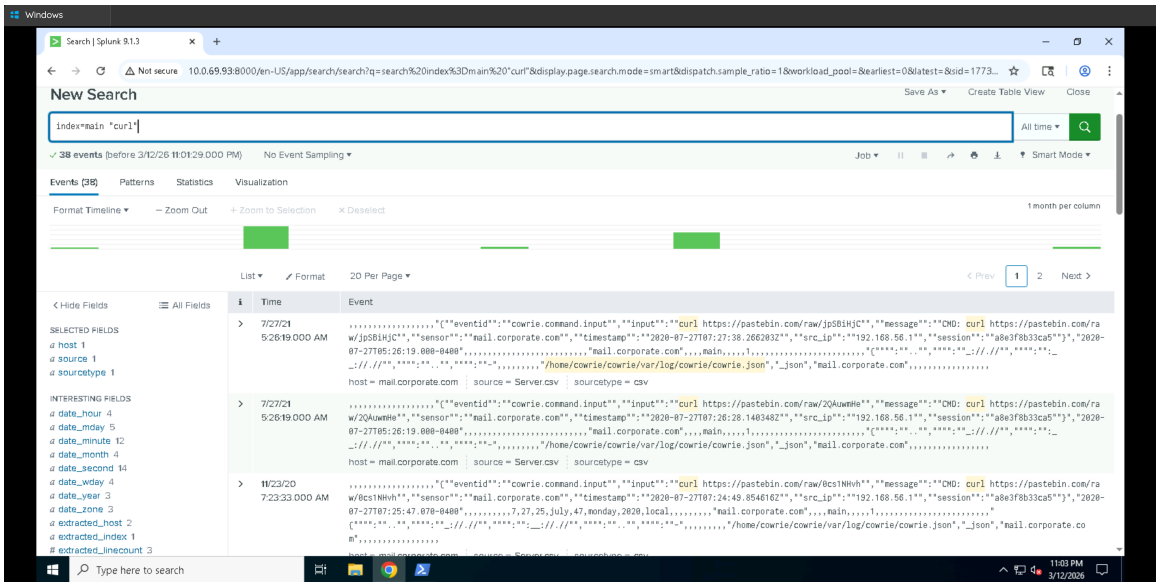

Figure 3- SIEM query revealing command-line payload retrieval activity.

This revealed multiple honeypot session events where external content was requested from pastebin.com, suggesting staged payload delivery.

3.4 Infrastructure Pivoting and IOC Correlation

Further hunting pivoted on the identified infrastructure indicator (Pastebin URL).

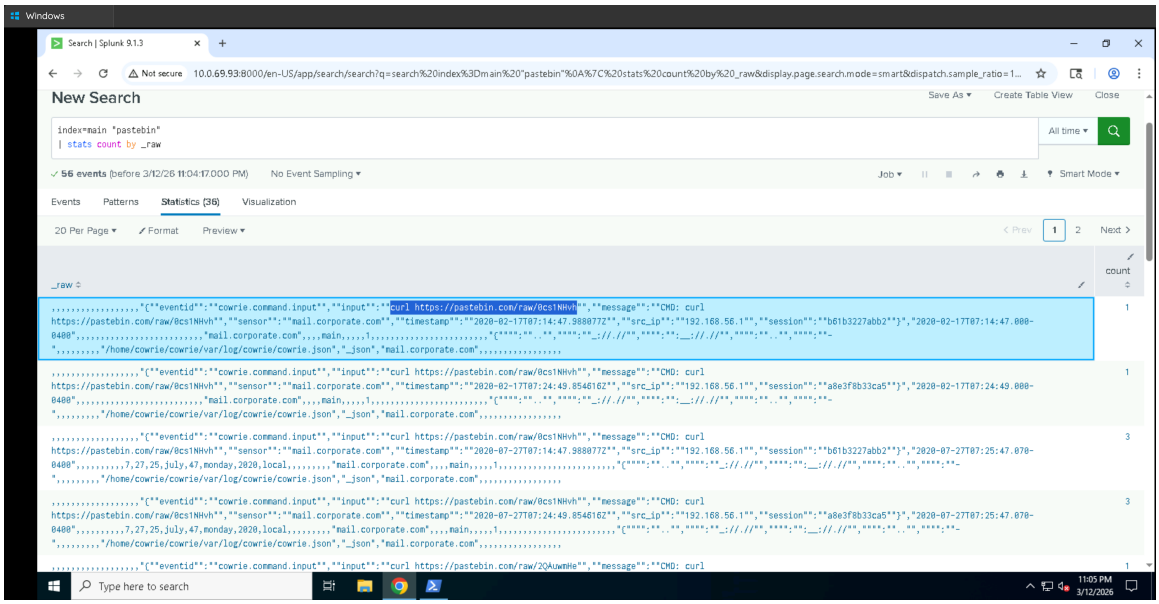

Figure 4- Correlation of repeated attacker infrastructure usage.

Repeated execution attempts indicated automated or persistent attacker behavior. In real environments, repeated outbound retrieval attempts may signal script deployment, malware installation routines, or command-and-control initialization.

3.5 Payload Retrieval and Obfuscation Analysis

The retrieved Pastebin content consisted of Base64-encoded data.

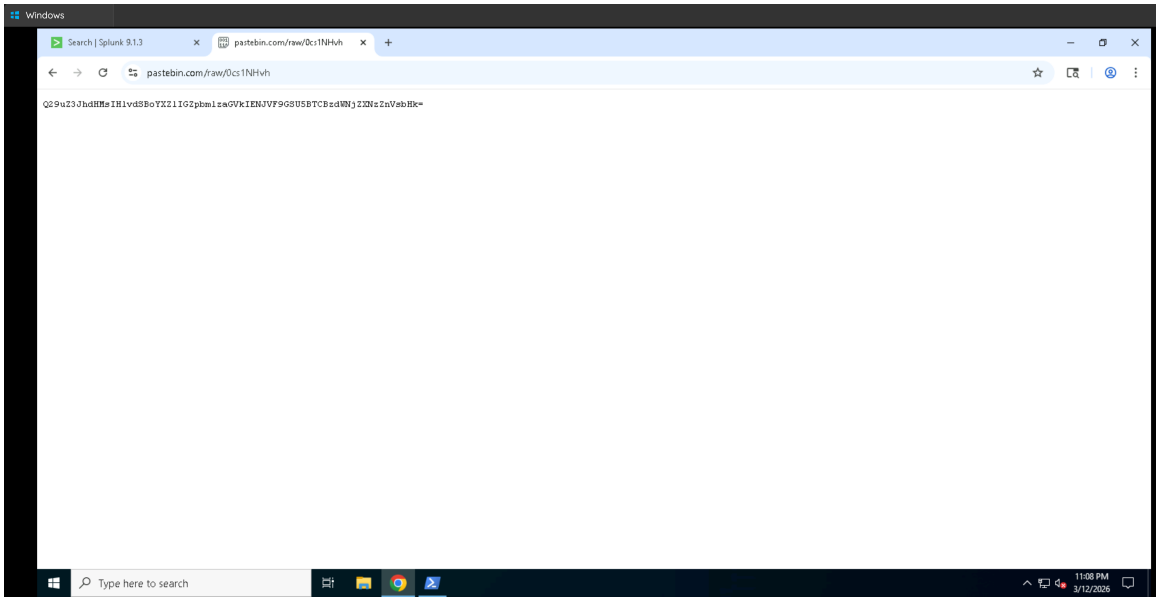

Figure 5- Encoded payload retrieved from attacker staging infrastructure.

Encoding techniques such as Base64 are frequently used by adversaries to evade simple detection mechanisms and conceal payload instructions. The artifact was decoded using CyberChef, confirming successful reconstruction of the simulated attack lifecycle.

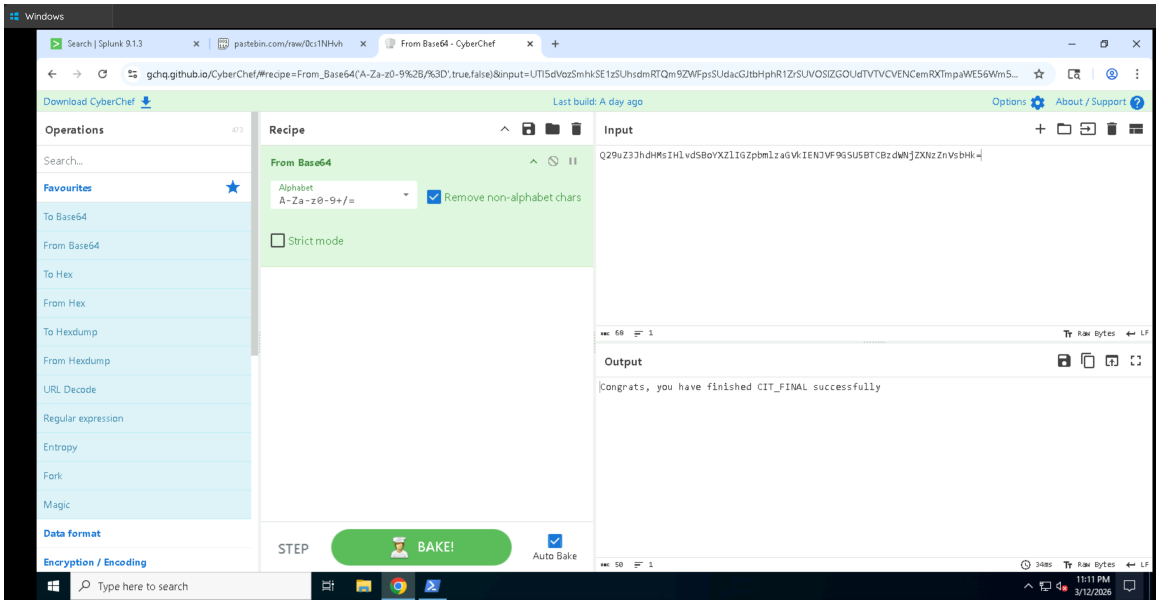

Figure 6- Successful decoding of attacker payload confirming simulated compromise lifecycle.

4. Key Findings

- Repeated command execution using curl within honeypot sessions.
- External payload staging via public infrastructure (pastebin.com).
- Log normalization weaknesses impacting detection efficiency.
- Encoded artifact delivery indicating adversary obfuscation techniques.
- Successful investigative pivoting enabling full threat reconstruction.

5. Indicators of Compromise

Source IP Address: 192.168.56.1
External Infrastructure: pastebin.com
Payload URL: https://pastebin.com/raw/0cs1NHvh
Technique: Command-line payload retrieval (Ingress Tool Transfer)
Sensor Host: mail.corporate.com (Cowrie Honeypot)

6. Risk Assessment

In production environments, similar activity could indicate malware deployment, persistence mechanisms, or lateral movement preparation. Outbound payload retrieval combined with encoded content delivery represents a high-risk behavioral indicator requiring immediate containment and forensic triage.

7. Recommendations

- Deploy detection rules for CLI-based file download activity.
- Improve SIEM field extraction and log normalization.
- Restrict outbound access to paste and file-sharing platforms.
- Implement network segmentation and honeypot isolation.
- Enforce secure credential management practices.

8. Conclusion

This investigation demonstrates the importance of behavioral threat hunting when structured detections are unavailable. The analyst successfully reconstructed attacker activity from limited telemetry, validated external staging infrastructure, and performed artifact decoding to confirm adversary intent. Such workflows closely mirror real SOC and DFIR operational scenarios.